

TriageLoop.ai

The deadline-aware operating layer
for emergency-department waiting rooms

A changing patient creates a changing clinical deadline.

We turn that need into an Action Window, test the live queue, and expose any shortfall as Clinical Slack.

PATIENT NEED



ACTION WINDOW

—

QUEUE ETA

=

CLINICAL SLACK

Triage is captured once. Risk keeps moving.

A category can stay frozen while both the patient and the queue change underneath it.

ARRIVAL SNAPSHOT

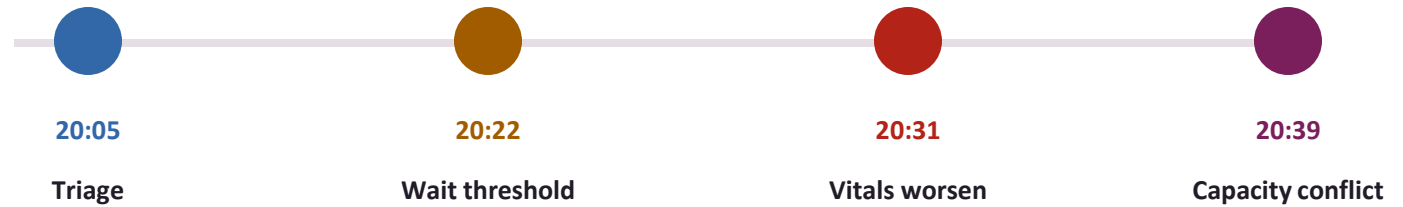
Level 3

Recorded at 20:05

**The category answers:
'How urgent at intake?'**

It does not continuously answer when the patient must be reassessed—or whether the ED can act in time.

THE SYSTEM KEEPS MOVING



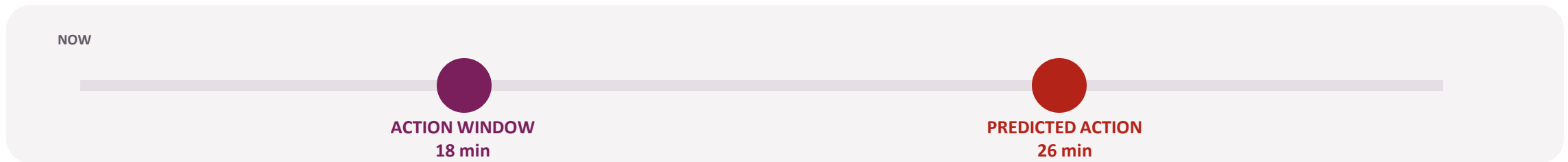
The operational question

Who is becoming unsafe to wait, by when must we act, and can the department meet that need?

A score describes risk. A deadline organizes action.

TriageLoop turns changing evidence into a conservative time requirement—then refuses to confuse ranking with capacity.

PATIENT-SPECIFIC ACTION WINDOW



Clinical Slack = $18 - 26 = -8$ minutes

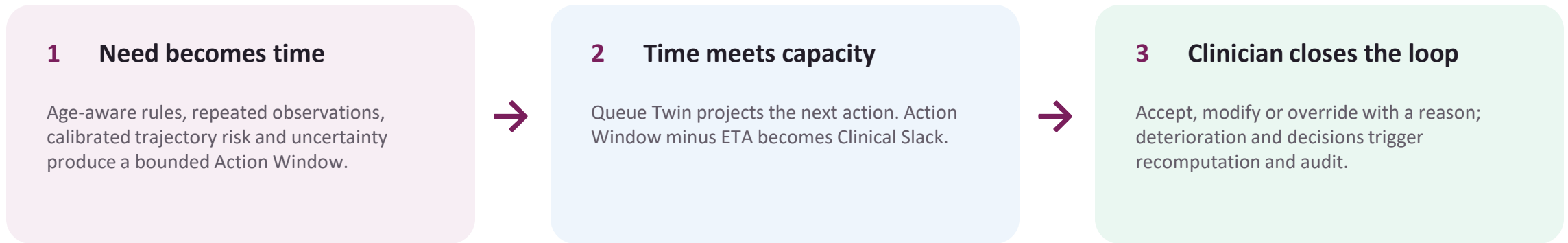
ILLUSTRATIVE EXAMPLE — DISTINCT FROM THE LIVE P-0009 CASE

CAPACITY CONFLICT

The clinical requirement stays fixed. The queue is declared infeasible.

One closed loop, three moves.

The model is bounded by deterministic rules; the queue cannot redefine clinical need; the clinician remains accountable.

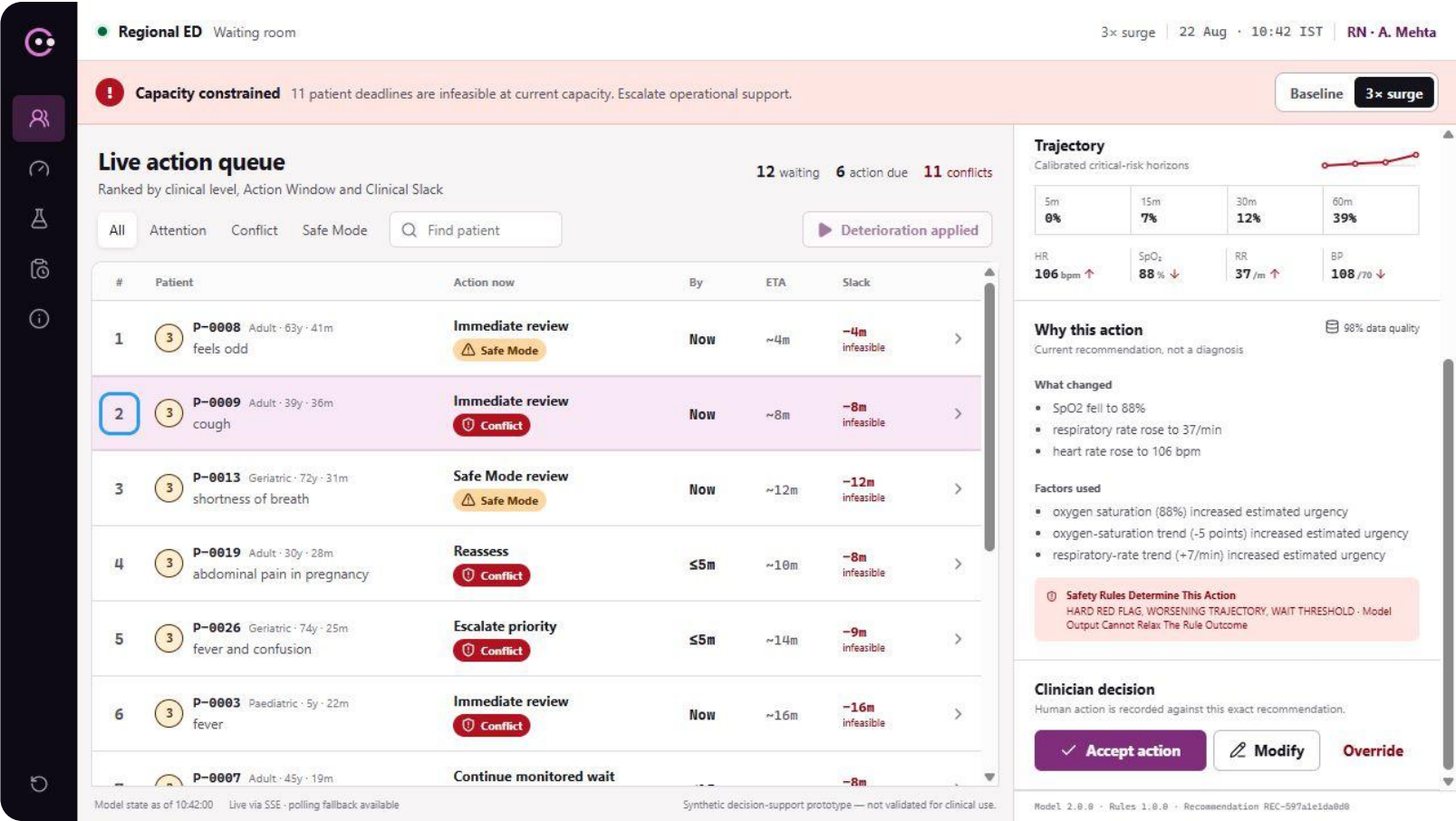


Every new observation can update the deadline. Every updated deadline recomputes the queue. Every action creates a receipt.

NO AUTONOMOUS DOWNGRADE

The Deadline Board is the product—not a decorative dashboard.

One quiet operations surface answers who, by when, why, whether capacity is feasible, and what the clinician decided.

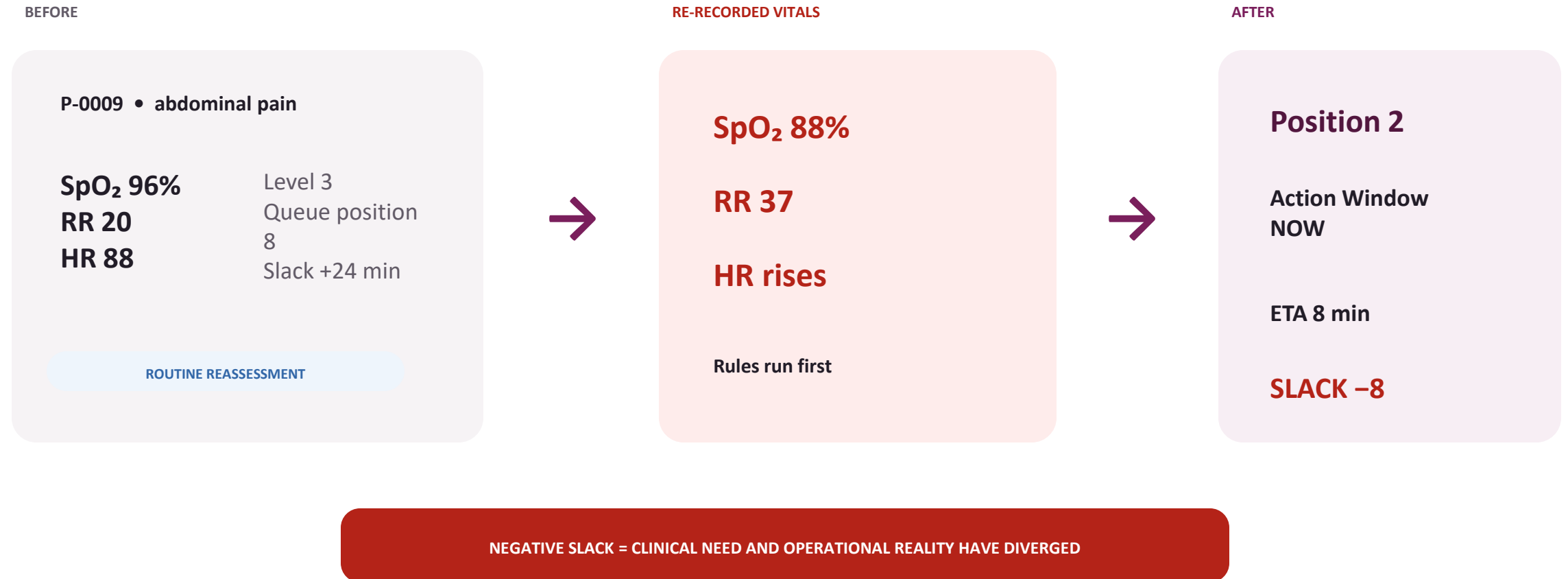


WHAT THE NURSE SEES

- 01 Action + deadline**
Lead with what must happen and by when.
- 02 Clinical Slack**
Capacity failure is explicit, not hidden in rank.
- 03 What changed**
Trajectory, rules and uncertainty beside the action.
- 04 Human decision**
Accept, modify or override with a reason.

One worsening observation changes the queue—and exposes capacity.

The system does not label the patient 'safe.' It shows the changed evidence, required action and infeasible response.



The model can sharpen the decision. It cannot overrule the safety rail.

Rules, uncertainty, queue feasibility and clinician authority are separate controls—not one opaque score.

- 01 INPUT + DATA QUALITY**
Freshness • missingness • plausibility • provenance
- 02 AGE-AWARE RULES FIRST**
Paediatric • adult • geriatric red flags
- 03 CALIBRATED TRAJECTORY**
Four horizons • recall-constrained operating point
- 04 UNCERTAINTY SHELL**
Conformal set • OOD • Safe Mode
- 05 ACTION WINDOW + QUEUE TWIN**
Deadline bounded by rules • capacity tested separately

LOCKED INVARIANTS

No autonomous downgrade

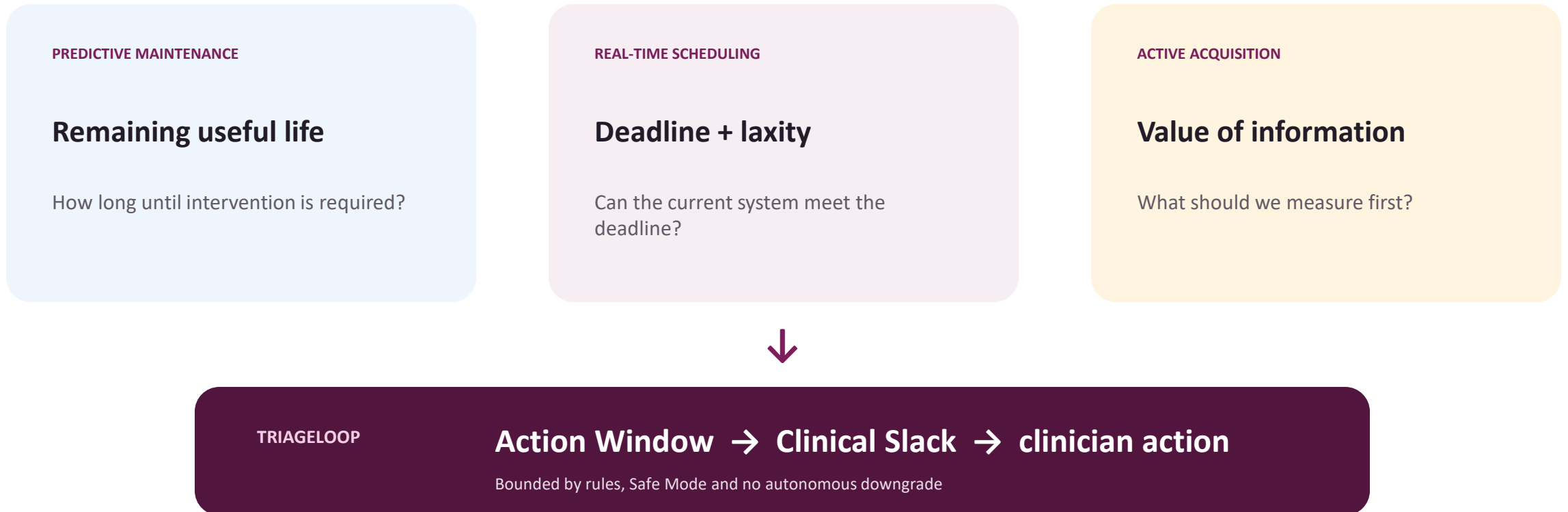
Hard red flags win

Uncertainty triggers review

Clinician owns the action

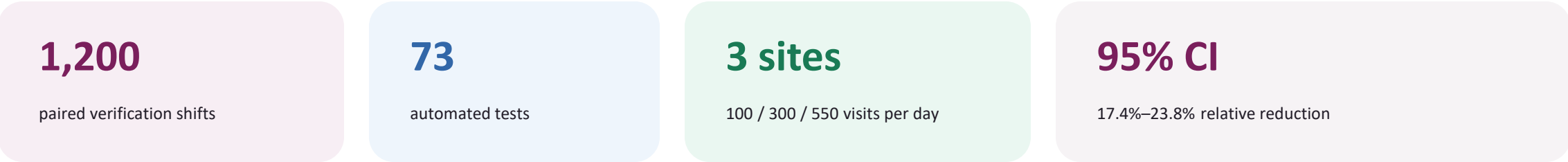
The wow factor is the recomposition—not an ‘AI triage’ claim.

Three proven ideas from other domains become one safety-oriented ED waiting-room control loop.



20.5% fewer Action Window misses under 3× surge.

The lead claim uses a stronger comparator: fixed 15-minute periodic re-triage, not a static arrival score.



ACTION WINDOW MISS RATE / 3× SURGE



Also: 14.0% fewer negative-Slack minutes • 15.7% faster signal-to-action • 14.4% lower low-acuity P90 wait

One hypothesis failed. The product boundary changed.

The single-observation Next Best Observation gate did not preserve operational critical recall.

87.5%

fewer observations requested

The efficiency signal looked promising.

BUT

−7.1 pp

operational critical recall

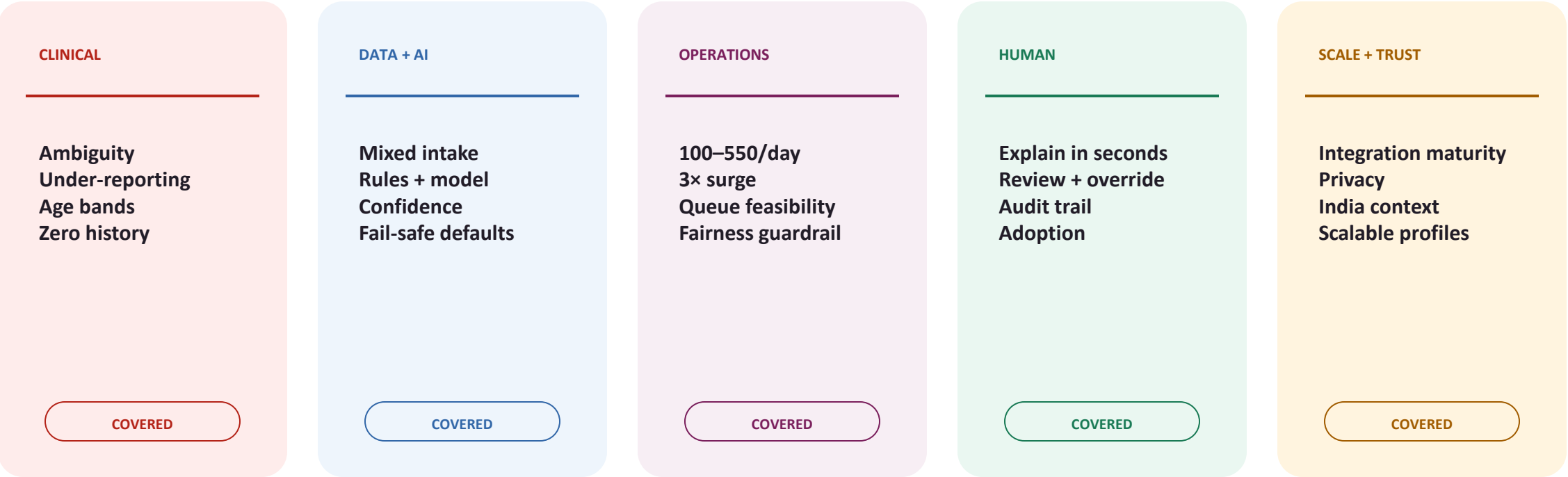
Predeclared limit: no more than −2 points.

LOCKED PRODUCT RESPONSE

**NBO is a first measurement suggestion—not a reassessment replacement.
Uncertainty → full reassessment or escalation.**

Every stated complexity has a visible response.

The proposal and prototype cover the whole brief; the center remains one coherent invention.



Detailed mapping: requirements traceability + proposal Appendix A

Fit the clinical workflow; do not ask the workflow to fit the model.

One consolidated signal, quiet unchanged rows and explicit human authority reduce alert fatigue and ambiguity.

01

TRIAGE NURSE

Who needs attention next—by when and why?

02

CHARGE NURSE

Where has clinical need exceeded available capacity?

03

QUALITY / SAFETY

Are rules, uncertainty, overrides and outcomes governed?

04

IT / DIGITAL

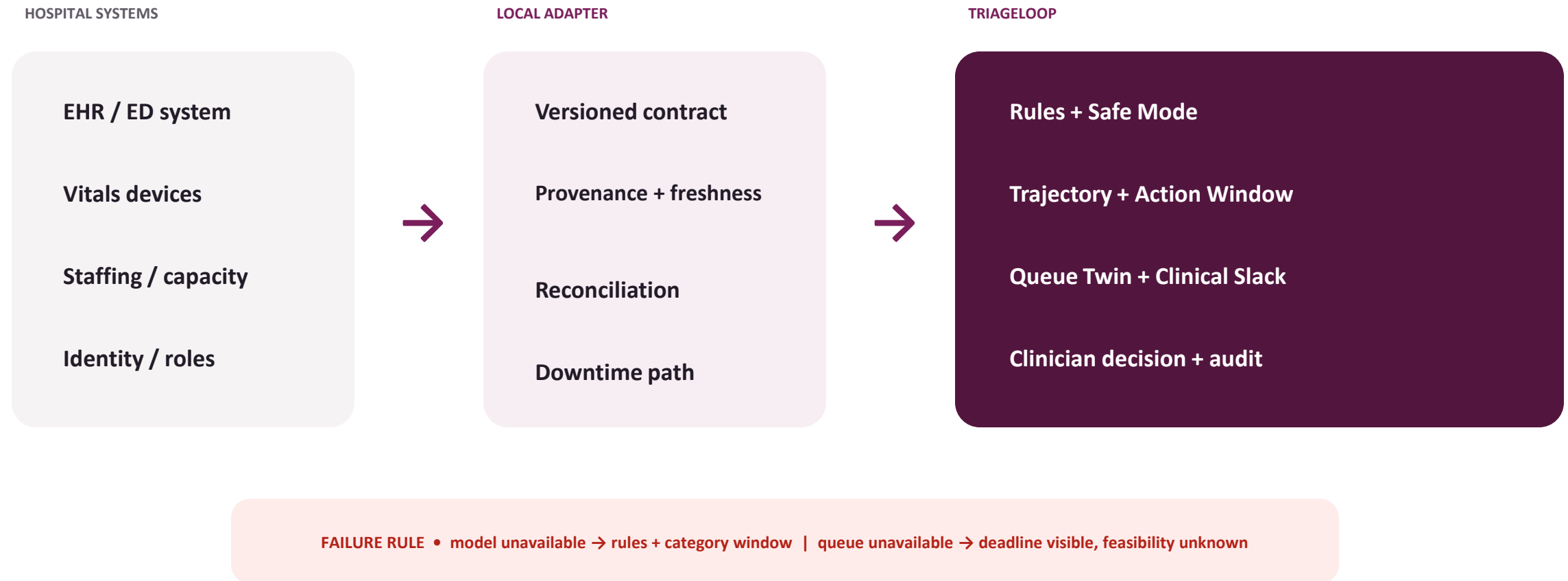
Can this integrate as a thin, observable layer?

CURRENT SYNTHETIC SURGE WORKLOAD • 0.86–1.12 alerts / waiting-patient-hour • 2.20–5.56 alerts / reassessment-nurse-hour

Shadow mode first • Community is the warning case • nurse task testing remains external

A thin operating layer—not an EHR replacement.

The clinical requirement survives partial failure; only feasibility becomes unknown when the queue service is unavailable.



Measure value before monetizing it.

No synthetic rupee-savings claim. A local pilot earns the right to scale through observable process and safety outcomes.

VALUE LEVERS

- Fewer missed action windows
- Faster deterioration review
- Less manual queue scanning
- Earlier capacity conflict visibility
- Auditable decisions

PILOT ECONOMICS

Annual attributable value

(avoidable process time × loaded staff cost)
+ operational exceptions avoided
+ audit / documentation efficiency
– integration, validation, change and operating cost

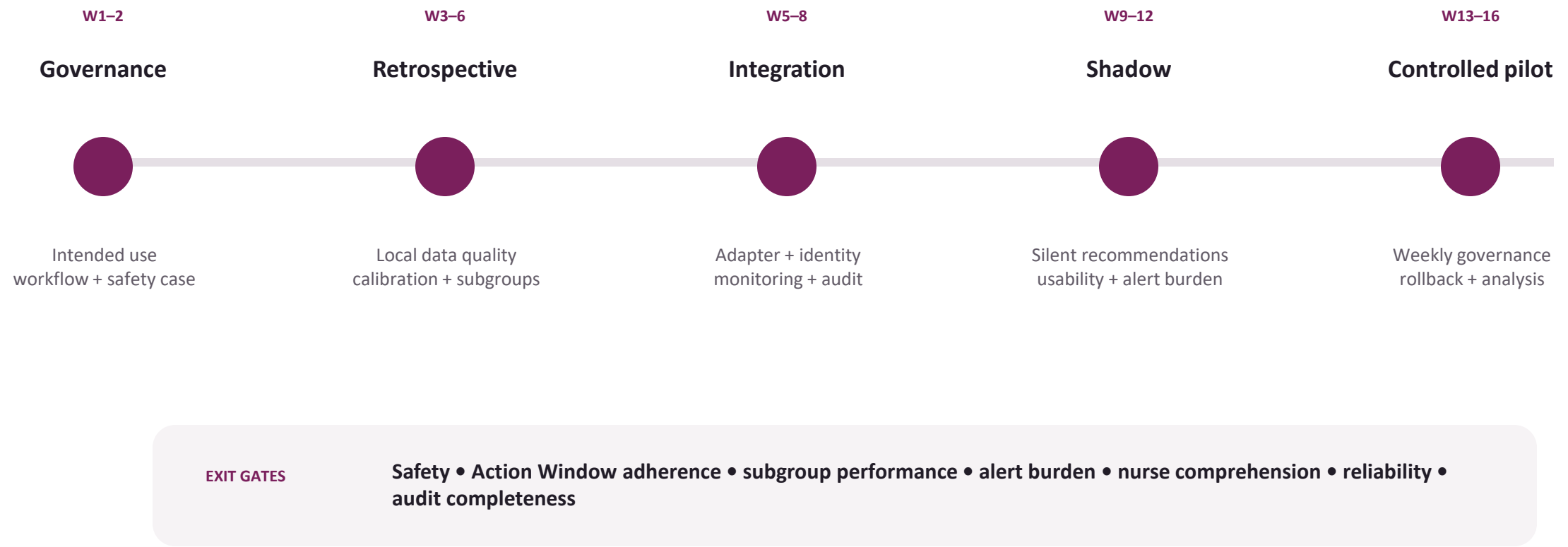
DO NOT MONETIZE

Patient harm avoided until independent clinical evidence supports the causal link.

Buyer path: retrospective validation → shadow pilot → controlled production → multi-site scale

A 16-week route from retrospective proof to controlled use.

Each phase has a go/no-go gate; safety or usability failure constrains the product instead of being narrated away.



Advance the deadline-aware model of triage.

Give every waiting patient a changing, explainable action deadline—and give the department an honest view of whether it can meet it.



When capacity is constrained, TriageLoop does not create false reassurance.

It shows where clinical need and operational reality have diverged—early enough for a clinician and charge nurse to act.

PROPOSED NEXT STEP • LOCAL RETROSPECTIVE + SHADOW PILOT

Verification design and claim boundaries

The evidence is reproducible, paired and intentionally bounded.

POPULATION	Seeded 10,000-encounter synthetic longitudinal generator + 28 curated edge cases
SITES	Community 100/day • Regional 300/day • Urban trauma 550/day
DEMAND	Baseline and 3× surge
MATRICES	TL-03: 1,800 shifts / 3 policies • TL-05: 1,200 shifts / 2 policies
POLICIES	Static/FIFO/TriageLoop in TL-03 • periodic re-triage/TriageLoop in TL-05
LEAD ENDPOINT	Action Window miss rate; paired shifts with bootstrap confidence interval
SENSITIVITY	Periodic comparator: 12.8% / 13.0% / 24.1% fewer misses at 10 / 20 / 30 min
BOUNDARY	Only 30 min exceeds 20%; post-verification analysis, not a retroactive TL-05 gate

NOT ESTABLISHED • real-world clinical accuracy, patient benefit, safe staffing, hospital savings or legal compliance

India governance baseline

Official policy references inform the design; deployment still requires hospital legal, security and clinical-governance review.

DPDP ACT 2023

Digital personal-data obligations and accountable processing baseline.

DPDP RULES 2025

Phased commencement; re-verify current obligations before deployment.

ABDM HDM POLICY

Privacy/security by design, consent concepts and federated architecture.

EHR STANDARDS INDIA

Interoperability, access control and audit expectations.

REFERENCE ≠ COMPLIANCE CERTIFICATION OR REGULATORY APPROVAL

Selected sources

Related work prevents novelty overclaim and supports the cross-domain design rationale.

Repeated vital signs

pubmed.ncbi.nlm.nih.gov/30005671

Prognosis-informed prioritisation

pubmed.ncbi.nlm.nih.gov/37318826

WHO triage tools

who.int/tools/triage

Predictive maintenance / RUL

ntrs.nasa.gov/citations/20140010623

MIMIC-IV demo

physionet.org/content/mimic-iv-demo/2.2

Dynamic deterioration modeling

pubmed.ncbi.nlm.nih.gov/41827109

Conformal ED ‘I don't know’ (PMID 40318497)

pubmed.ncbi.nlm.nih.gov/40318497

Deadline / laxity scheduling

people.eecs.berkeley.edu/.../lec11-Scheduling.pdf

Active feature acquisition

research.ibm.com/projects/active-feature-acquisition

Official competition page

unstop.com/.../accenture-innovation-challenge-2026...
